Greedy Problem Solving

- Viraj Chandra

Problem 1

Find Original Array From Doubled Array:

https://leetcode.com/problems/find-original-array-from-doubled-array/description/

original = $\{ a_1 \ a_2 \ a_3 \}$

$$\text{changed} = \{ a_1 \ a_2 \ a_3 \ \underset{}{\overset{a_4}{2 \times a_1}} \ \overset{a_5}{2 \times a_2} \ \overset{a_6}{2 \times a_3} \}$$

input ?

find this

$\rightarrow \{ \ \}$

$\rightarrow$ original

Base Case

→ n ↓

→ n X 2 → even

↳ odd - { }

$a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5 \quad a_6$

$x \rightarrow 2x$

$x/2 \quad \times$

smallest

$\cancel{a_1}$ $\cancel{a_2}$ $\cancel{a_3}$ $\cancel{a_4}$ $\cancel{a_5}$ $\cancel{a_6}$

$2\times a_2$ $2\times a_1$ $2\times a_5$

$\{ \quad \}$

1 3 4 2 6 8

1 4 3 2 8 6

Problem 2

Bracket Coloring:

https://codeforces.com/problemset/problem/1837/D

(	(	(	)	)	)	)	(
1	1	1	2	1	1	1	2
1	1	1	1	1	1	2	2

1 → ((())) → RBS

2 →)(→ () → RBS

① Valid parenthesis → stack

RBS

① → '('s = ')'s

② → at any given moment of time from left, the no. of '(' >= no of ')'

② RRBS

'('s = ')'

')'s $\geq$ '('s

③

$$RBS_1 + RBS_2$$

$$\Rightarrow RBS$$

④ $RRBS_1 + RRBS_2$

$$\Rightarrow RRBS$$

$S_1 = S_2 = S$

$S_1 + S_2$

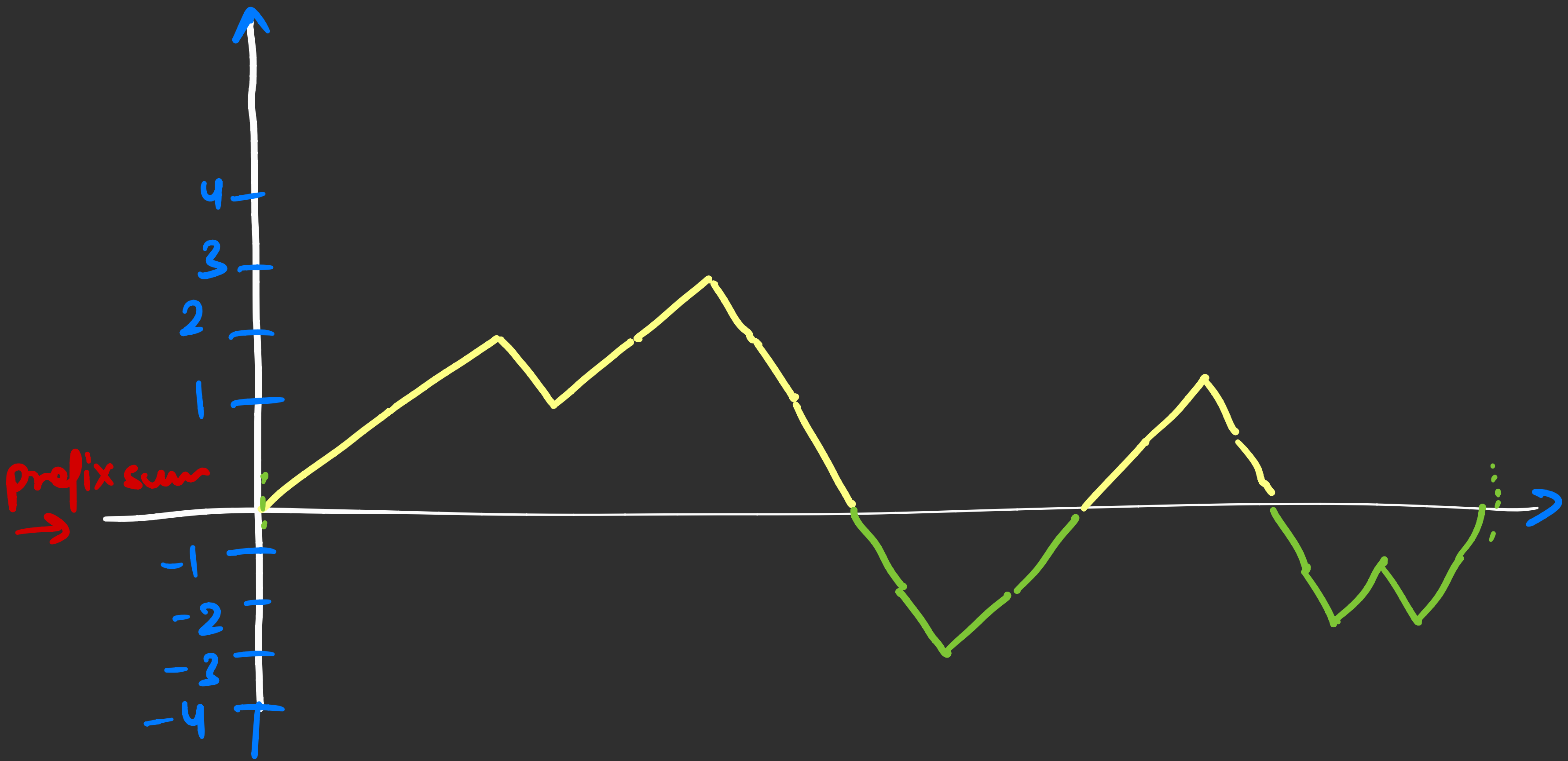
$S + S = n$

$2S = n$

$S = n/2$

n is odd "-1"

'(' $\neq$ ')' "-1"



prefix sum
4
3
2
1
-1
-2
-3
-4

$$\begin{array}{cccccc} + & + & + & - & - & - \\ (& (& (&) &) &) \\ 1 & 2 & 3 & 2 & 1 & 0 \end{array}$$

RBS

ob $\rightarrow$ +1

cb $\rightarrow$ -1

s

Problem 3

Arranging The Sheep:

https://codeforces.com/problemset/problem/1520/E

Proof:

https://math.stackexchange.com/questions/113270/the-median-minimizes-the-sum-of-absolute-deviations-the-ell-1-norm

* . * . . . * . * *

$O(n^2)$

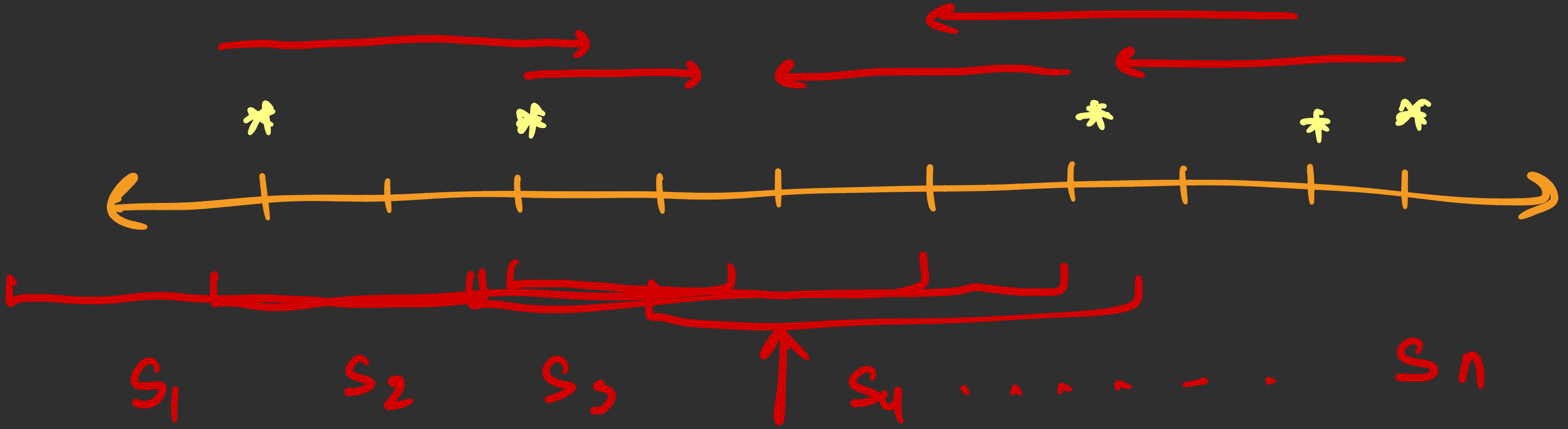
#1 → m_1

#2 → m_2

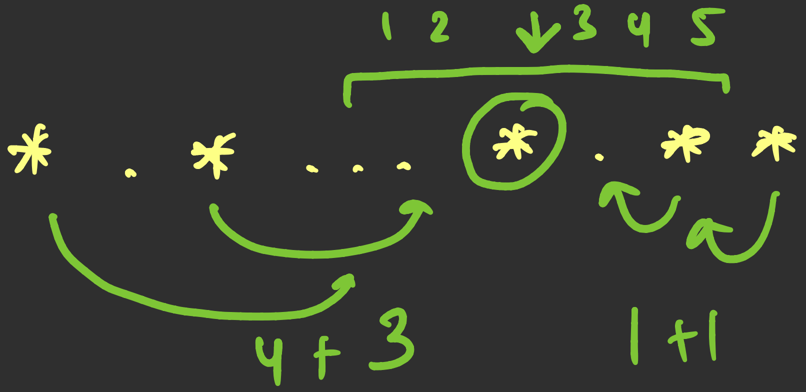
#3 → m_3

#4 ⋮

→ minimum



Sum = all red lengths



$\Rightarrow 9$

find the minimum deviation of all values of a[i]

$S_1 > S_2 > S_3 > S_4 > S_5 \ldots S_m < S_{m+1} < S_{m+2} \ldots$

↳ median of the arr